



Dynamic Small Signals Self-Synchronizing System

Introduction

- Traditional synchronization methods and flaws:
 - **Phase-Locked Loops & Delay-Locked Loops:** Use complex feedback loop; Modify original signals; Trigger different jitters.
 - **Length Matching:** Match all wires to the longest wire; Create more interference
- Goal: To to build a **low-cost, simple-logic** prototype to **self-synchronize multiple low-amplitude input signals** while **preserving the original signal integrity**.
- Expected Outcome:
 - **Synchronization Precision:** delay < 50 ns.
 - **Low noise output.**
 - **Cost <\$5.00.**

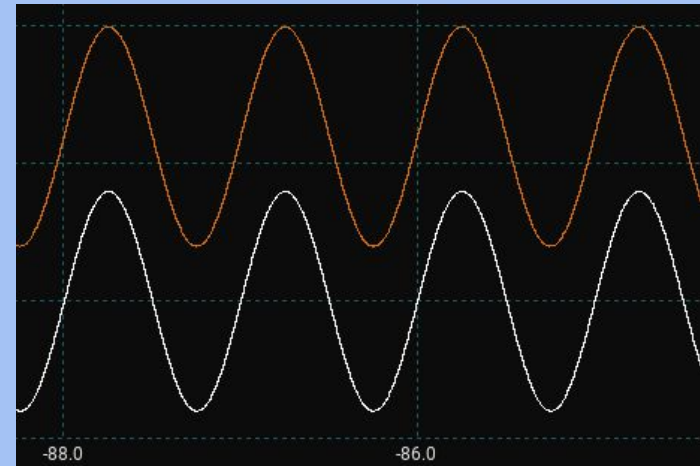


Figure 1: Synchronized Signals.
Image created by student using PicoScope 7 T&M, 2026

Methods

Design:

- Research and confirm technical solution
- Design schematic
- Build prototype on breadboard

Testing:

- Measure waves by oscilloscope.
- Use the built-in delay measurement tool to measure delay



Figure 2: Example of testing with measuring “0”s delay of output signals. The channel on the top is the input and the channel at the bottom is the output. Image created by student using PicoScope 7 T&M, 2026.

Methods

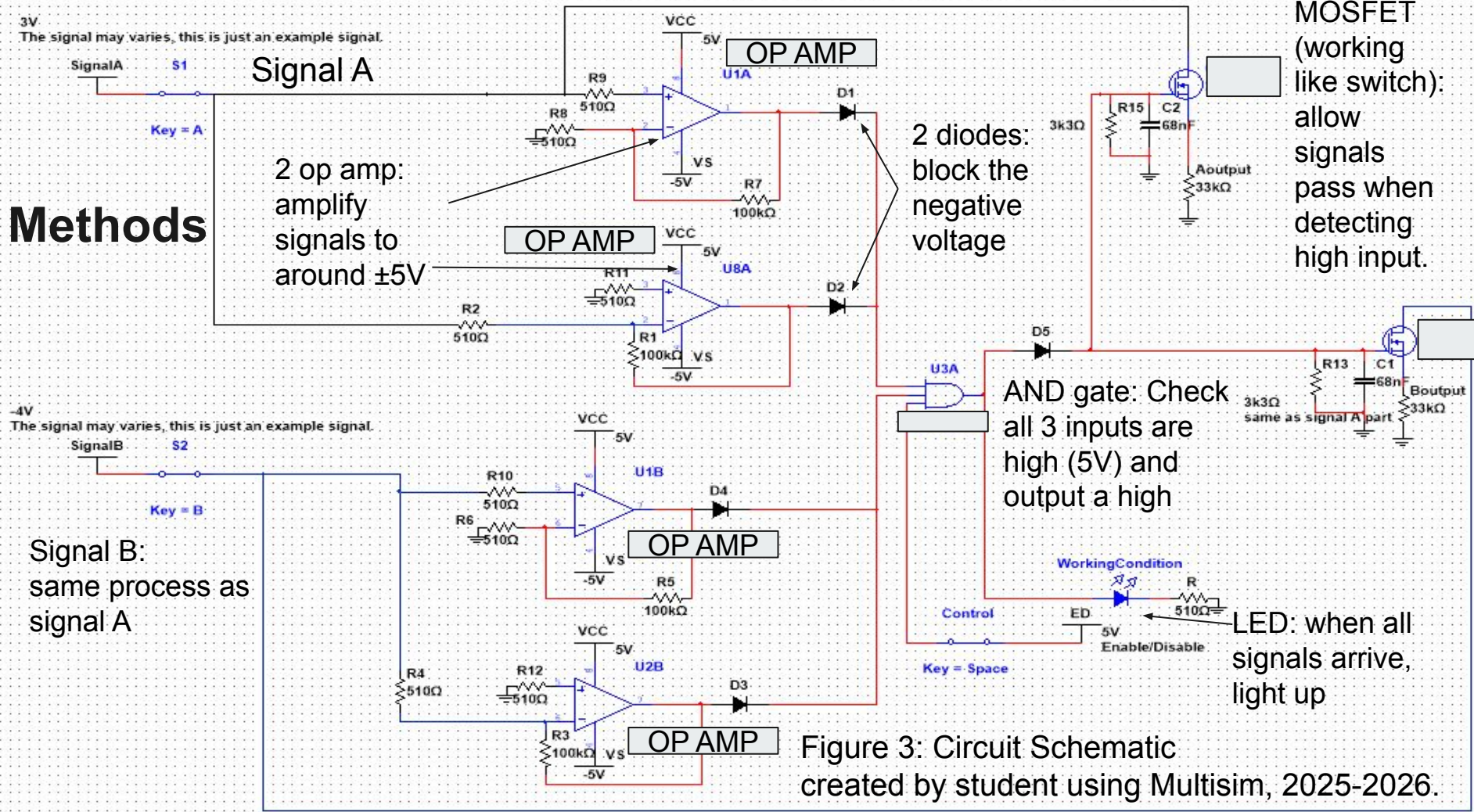


Figure 4 & 5: Analyzed Datasets for sine waves.
Graph created by student using Google Sheets, 2026

Results

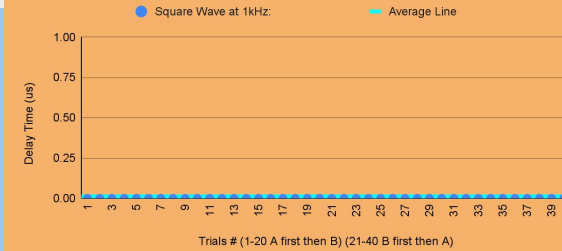
Note: The “0”s are due to equipment accuracy to measure <4ns delay.

- My prototype met my engineering goal by:
 - 96.67% of trials <50 ns delay**
 - Low cost: **\$2.97**
 - No modification of the original signal
- Testing (120 trials):
 - Measuring “0s” delays.
 - Square Waves** (80 trials):
 - Average delay (98.75%<4ns):
 - 1kHz: <=4ns.
 - 3kHz: <=29 ns
 - Sine Waves (40 trials):**
 - Average delay (92.5%<=4ns)
 - 10kHz: 0.10 μ s

Figure 6 Analyzed
Datasets for sine wave at
10kHz.
Graph Created By Student
using Google Sheets,
2026

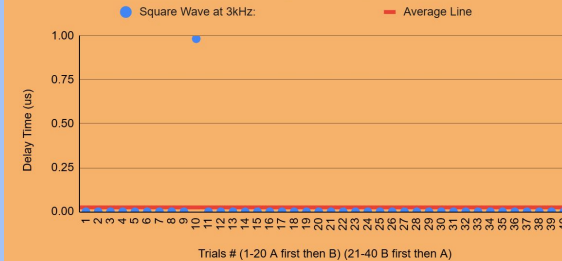
Square Wave at 1kHz Delay Measurement Between
Synchronized Signals

Standard Deviation = 0 us (0ns) Average = 0 us (0 ns)



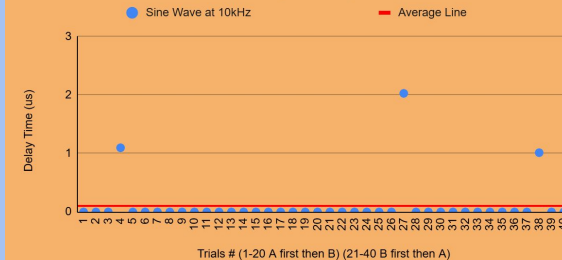
Square Wave at 3kHz Delay Measurement Between
Synchronized Signals

Standard Deviation: 0.16 us (160 ns) Average: 0.025 us (25 ns)



Sine Wave at 10kHz Delay Measurement Between
Synchronized Signals

Standard Deviation: 0.39 us (390ns) Average: 0.10 us (100ns)



Discussion



Performance of the Circuit:

- **Square Waves ($\leq 3\text{kHz}$):**
 - **98.5% trials ns-level delay**
 - **Average Delay $\leq 16.5\text{ns}$**
- **Continuous Signals (10kHz Sine Waves):**
 - **92.75% trials ns-level delay**

Theoretical Outcomes:

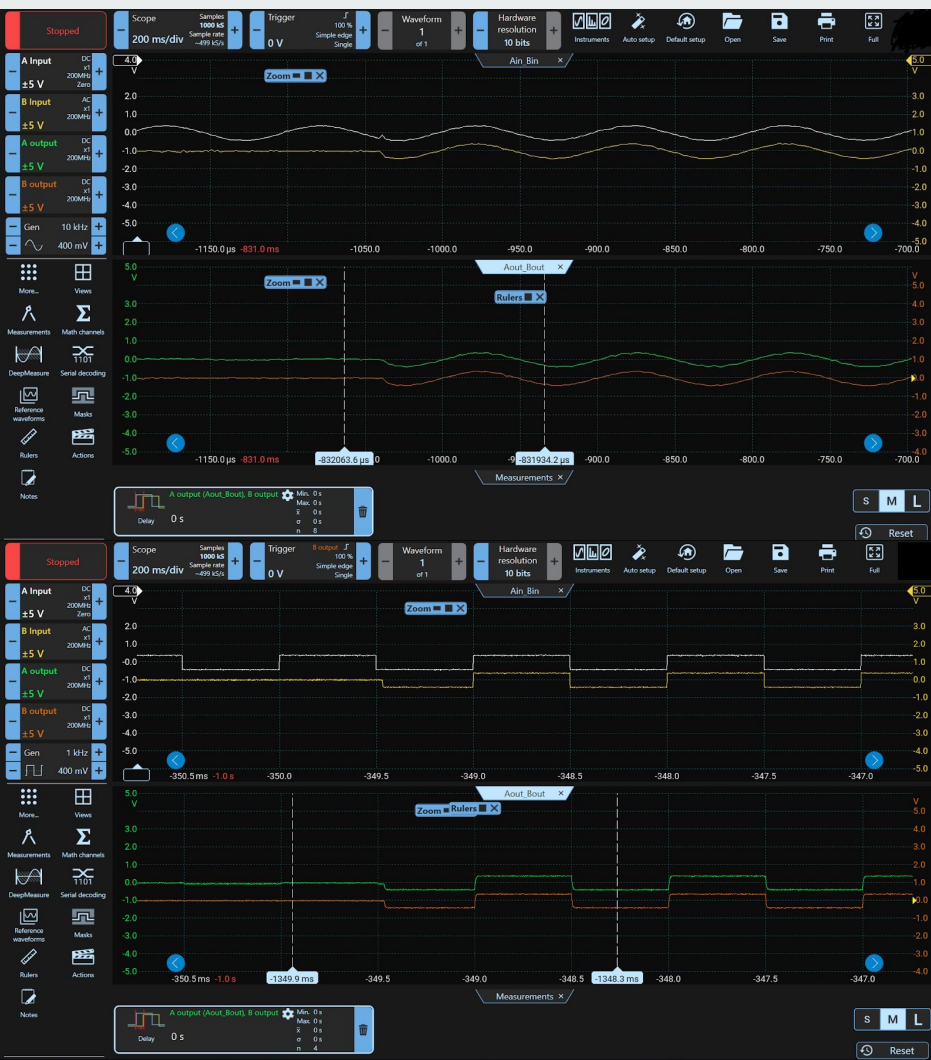
- **Maximum delay: $\leq 10\text{ ns}$** (due to switches/MOSFET turn-on time difference)
- **Maximum Bandwidth: 60kHz** ($f_{\text{max}} = 12\text{MHz}/200 = 60\text{kHz}$)

Figure 7 & 8: Well Performance Frequency Levels
Image created by student using PicoScope 7 T&M, 2026

Discussion

Technical Challenges & Mitigations

- **Noise:** Noise in synchronized signals
 - Solved by extending RC discharging time
- **Negative Signal Synchronization**
 - Solved by using 2 Op-Amps (coefficient ± 200)
- **Bandwidth**
 - **10kHz** (continuous signals) & **3kHz** (fast signals)
 - **Cause:** High **parasitic effects** from the breadboard
 - **Solutions** (in the future):
 - Design a **PCB** (minimize parasitic effects)
 - Upgrade to higher bandwidth & speed components



Discussion

Advancements

- No Modification of the Original Signal
- Automatic Synchronization
- Small-Signal Sensitivity (20mv).
- Low Noise
- Costs:
 - \$2.97 (producing 1).
 - <\$0.5 per board (producing 1000)

Figure 10: Screenshot of an important portion of BOM about the cost of producing 1000 boards (**\$345 total for materials; \$0.345 per board**). Image created by student using EasyEDA, LCSC, and exported to Excel, 2026

EST TOTAL :\$2.97			
Diode Resistor AND Gate MOSFET Capacitor LED OP-AMP	Order Qty.	Unit Price(USD)	Ext.Price(USD)
	100	0.0031	0.31
	100	0.0015	0.15
	1	0.4130	0.41
	50	0.0118	0.59
	100	0.0005	0.05
	50	0.0151	0.76
	5	0.1394	0.70

Figure 9: Screenshot of an important portion of BOM (Bill of Materials) about the minimum amount to buy and the cost (**\$2.97**) of building **1 prototype**.

Image created by student using EasyEDA, LCSC, and exported to Excel, 2026

EST TOTAL :\$345.00			
Diode Resistor AND Gate MOSFET Capacitor LED OP-AMP	Order Qty.	Unit Price(USD)	Ext.Price(USD)
	5000	0.0027	13.50
	17000	0.0013	22.10
	1000	0.1364	136.40
	2000	0.0078	15.60
	2000	0.0004	0.80
	1000	0.0118	11.80
	2000	0.0724	144.80

Conclusion



- **My project turned out as expected in these:**
 - Synchronization precision
 - Low-cost
 - Automatic-synchronization
 - No modification of the original signal
- **Benefits**
 - Allow signals pass to device only during all valid
 - Release controller calculation capability while waiting for valid signals
 - Enhance controller real time response
- **Application Fields**
 - Automotive sensors
 - Industrial automation
 - Medical sensors

References

I am thankful for my dad's support and encouragement. He influenced me to think by myself critically and discuss about my thinking when facing problems.
I am thankful for my AP Physics 1 teacher, who answer my questions about how deep my project should aim for and how to manage my time.
I am thankful for my mentor for double-checking my designs.

Brown, J., & Nicholas, K. (2021). Design Considerations for Multiple Wide Bandwidth Delta- Sigma ADCs in Simultaneous-Sampling Systems. In ti.com. Texas Instruments. Retrieved from Texas Instruments website: <https://www.ti.com/lit/an/sbaa520a/sbaa520a.pdf>

Halio, J. (n.d.). Rotational Measurements with Resolvers energize the input phase with TSB582. Retrieved February 27, 2026, from st.com website: <https://www.st.com/content/dam/specialevents-assets/dc22/pdf/room-2-7-stdevcon22-session-rotational-measurements-w-resolvers.pdf>

Introduction to Piezoelectric Accelerometers. (n.d.). Retrieved February 27, 2026, from www.pcb.com website: <https://www.pcb.com/resources/technical-information/introduction-to-accelerometers>

Microchip. (2020). 2N7000 Features • Free from Secondary Breakdown • Low Power Drive Requirement • Ease of Paralleling • Low C ISS and Fast Switching Speeds • Excellent Thermal Stability • Integral Source-Drain Diode • High Input Impedance and High Gain General Description N-Channel Enhancement-Mode Vertical DMOS FET. In <https://microchip.com/>. Retrieved from <https://ww1.microchip.com/downloads/en/DeviceDoc/2N7000-N-Channel-Enhancement-Mode-Vertical-DMOS-FET-Data-Sheet-20005695A.pdf>

NXP Semiconductors. (2012). Using Decoupled Parallel Mode for Safety Applications [White Paper]. Retrieved from <https://www.nxp.jp/docs/en/white-paper/MPC564XLWP.pdf>

Obertino, M. (2023). Upgrade of the E/E Architecture of an electric test vehicle with drive-by-wire component. In <https://webthesis.biblio.polito.it>. Karlsruhe Institute of Technology. Retrieved from Karlsruhe Institute of Technology website: <https://webthesis.biblio.polito.it/29784/1/tesi.pdf>

P2138 code: throttle/pedal position sensor. (2025, November 21). Retrieved February 27, 2026, from foxwelldiag.com website: <https://www.foxwelldiag.com/blogs/car-diagnostic/p2138-code-throttle-pedal-position-sensor>

Soundarapandian, K., & Berarducci, M. (2009). Application Report Analog Front-End Design for ECG Systems Using Delta-Sigma ADCs. In ti.com. Texas Instruments. Retrieved from Texas Instruments website: <https://www.ti.com/lit/an/sbaa160a/sbaa160a.pdf>

Texas Instruments. (1979). NE5532, NE5532A, SA5532, SA5532A NE5532x, SA5532x Dual Low-Noise Operational Amplifiers 1 Features 3 Description. In ti.com. Retrieved from <https://www.ti.com/lit/ds/symlink/ne5532.pdf>

Won, H., Yoon, T., Han, J., Lee, J.-Y., Yoon, J.-H., Kim, T., ... Bae, H.-M. (2015). A 0.87 W Transceiver IC for 100 Gigabit Ethernet in 40 nm CMOS. IEEE Journal of Solid-State Circuits, 50(2), 399–413. <https://doi.org/10.1109/jssc.2014.2369494>